

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Concept for a Miniaturized Confocal Plane Mass Spectrometer using Micromachined Detector Array

Authors & Institutes:

- Adi Scheidemann — *Intelligent Ion, Inc.*
- Mahadeva Sinha — *Jet Propulsion Laboratory*
- Bruce Darling — *University of Washington*

ABSTRACT

The concept of a miniaturized Mattauch-Herzog mass spectrometer will be described; scaling laws governing miniaturizing magnetic sector mass spectrometer will be addresses and the pros-cons of the Mattauch-Herzog concept versus quads and ToF will be discussed. A DRIE based Faraday cup detector array will be compared to and an electro optical ion detector.